SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO4	**Assessment:**	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	**Total Marks:**	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	y.hema sai lahari	**Reg. No.:**	192311446
Date:	25-08-2026	**Duration:**	60 minutes

Code of Conduct

I y.hema sai lahari(192311446) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: y.hema sai lahari

Annexure A – Test Case Design Assignment

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, **design and document comprehensive test cases** to verify the correctness, functionality, reliability, and usability of the proposed system.

Your test case design should include:

1. Identify the **functional and non-functional requirements** to be tested.

2. Identify the **test scenarios** for the assigned problem statement.
3. Prepare detailed **test cases** covering normal, boundary, invalid, and exceptional conditions.
4. Include **Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.
5. Apply appropriate **test design techniques** such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
6. Ensure that the test cases provide adequate **requirement and functional coverage** for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15 %)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal,	Covers most important conditions	Covers basic conditions but misses several	Test cases are very limited or incomplete

	boundary, invalid, and exceptional conditions	with minor gaps	important cases	
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation &	Well-structured,	Well-organized	Adequately organized but	Poorly organized, unclear, or

Presentation(10%)	clear, professional, consistent, and easy to understand	with minor presentation issues	has some clarity/formatting issues	incomplete documentation

Criteria	**Mark**
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

ANNEXURE A – TEST CASE DESIGN DOCUMENT

TEMPLE DONATION AND EVENT MANAGEMENT SYSTEM

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1. Introduction

The **Temple Donation and Event Management System** is a web-based application designed to manage temple donations, devotees, events, and related administrative activities efficiently. The system allows devotees to register, make donations, view donation details, register for temple events, and receive confirmation. Administrators can manage devotees, donations, events, and generate reports.

This document presents comprehensive test cases to verify the **functionality, reliability, usability, security, and performance** of the proposed system.

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2. Objectives of Testing

The main objectives of testing are:

- To verify that all system functions work according to the requirements.
- To ensure that donation transactions are recorded correctly.
- To verify secure login and user authentication.
- To verify event creation, modification, and registration.
- To ensure accurate donation receipts and reports.
- To identify invalid and exceptional conditions.
- To verify that the system is easy to use.
- To verify that unauthorized users cannot access administrative functions.
- To ensure data integrity and reliability.

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3. Functional Requirements to be Tested

Requirement ID	Functional Requirement
FR01	User registration and login
FR02	User profile management
FR03	Temple information viewing
FR04	Donation category selection
FR05	Donation amount entry
FR06	Online donation/payment processing
FR07	Donation confirmation and receipt generation
FR08	Donation history viewing
FR09	Event creation and management
FR10	Event registration
FR11	Event registration confirmation
FR12	Admin dashboard
FR13	Devotee/user management
FR14	Donation record management

Requirement ID	Functional Requirement
FR15	Event record management
FR16	Report generation
FR17	Search and filter functionality
FR18	Logout functionality

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4. Non-Functional Requirements to be Tested

Requirement ID	Non-Functional Requirement
NFR01	System should provide acceptable response time.
NFR02	User data and donation information should be secure.
NFR03	System should be easy to understand and use.
NFR04	System should work correctly on commonly used browsers.
NFR05	System should maintain data accuracy and consistency.
NFR06	System should handle multiple users appropriately.
NFR07	System should recover appropriately from unexpected errors.
NFR08	System should provide reliable donation and event records.

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5. Test Scenarios

The following major test scenarios are identified:

1. Verify user registration.
2. Verify login with valid and invalid credentials.
3. Verify logout.
4. Verify user profile management.
5. Verify donation category selection.
6. Verify donation amount validation.
7. Verify successful donation.

8. Verify failed or cancelled payment.
9. Verify donation receipt generation.
10. Verify donation history.
11. Verify event creation by administrator.
12. Verify event modification and deletion.
13. Verify event registration by devotees.
14. Verify duplicate event registration.
15. Verify event capacity handling.
16. Verify admin access control.
17. Verify donation search and filtering.
18. Verify report generation.
19. Verify invalid input handling.
20. Verify system performance and usability.

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6. Test Design Techniques

6.1 Equivalence Partitioning

Input data is divided into valid and invalid groups.

Example – Donation Amount

- Valid partition: Amount greater than or equal to minimum allowed amount.
- Invalid partition: Amount less than minimum amount.
- Invalid partition: Zero.
- Invalid partition: Negative value.
- Invalid partition: Alphabetic/special-character input.

6.2 Boundary Value Analysis

Values at the minimum and maximum limits are tested.

Example:

If the minimum donation amount is ₹10:

- ₹9 – Invalid

- ₹10 – Valid
- ₹11 – Valid

If the maximum allowed amount is ₹1,00,000:

- ₹99,999 – Valid
- ₹1,00,000 – Valid
- ₹1,00,001 – Invalid

6.3 Decision Table Testing

Different combinations of conditions are tested.

Condition	Case 1	Case 2	Case 3
Valid login	Yes	Yes	No
Admin role	Yes	No	No
Admin dashboard access	Allowed	Denied	Denied

6.4 State Transition Testing

The donation/payment process can be represented using states:

Initiated → Payment Processing → Successful → Receipt Generated

or

Initiated → Payment Processing → Failed/Cancelled

Each possible transition should be tested.

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7. Detailed Test Cases

Module 1 – User Registration and Login

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC001	Register with valid details	Open registration → Enter details → Submit	Name: Hema, Email: hema@gmail.com, Password: Hema@123	Account should be created successfully	To be filled	Pass/Fail

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC002	Register with empty fields	Open registration → Leave fields empty → Submit	Blank fields	Appropriate validation messages should be displayed	To be filled	Pass/Fail
TC003	Register with invalid email	Enter registration details with invalid email	hema@	System should reject invalid email format	To be filled	Pass/Fail
TC004	Register with existing email	Enter an already registered email	existing@gmail.com	System should display account already exists message	To be filled	Pass/Fail
TC005	Login with valid credentials	Enter email and password → Click Login	Valid email/password	User should be successfully logged in	To be filled	Pass/Fail
TC006	Login with wrong password	Enter valid email and wrong password	hema@gmail.com / Wrong123	Login should be denied	To be filled	Pass/Fail
TC007	Login with unregistered email	Enter unregistered email and password	unknown@gmail.com	Login should fail with appropriate message	To be filled	Pass/Fail
TC008	Logout	Login → Click Logout	Valid user account	User should be logged out and redirected to	To be filled	Pass/Fail

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
				login/home page		

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8. Module 2 – Donation Management

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC009	Open donation page	Login → Select Donation	N/A	Donation page should be displayed	To be filled	Pass/Fail
TC010	Select valid donation category	Open donation page → Select category	Annadanam	Selected category should be accepted	To be filled	Pass/Fail
TC011	Enter valid donation amount	Enter amount → Continue	₹500	Amount should be accepted	To be filled	Pass/Fail
TC012	Enter minimum donation amount	Enter minimum permitted amount	₹10	Amount should be accepted	To be filled	Pass/Fail
TC013	Enter amount below minimum	Enter amount below minimum	₹9	Validation message should be displayed	To be filled	Pass/Fail
TC014	Enter zero donation	Enter ₹0	₹0	System should reject the amount	To be filled	Pass/Fail
TC015	Enter negative donation	Enter negative amount	-₹100	System should reject the amount	To be filled	Pass/Fail

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC016	Enter alphabetic amount	Enter letters in amount field	ABC	System should reject invalid input	To be filled	Pass/Fail
TC017	Successful donation	Select category → Enter amount → Make payment	₹500, valid payment details	Payment should be successful and donation should be recorded	To be filled	Pass/Fail
TC018	Failed payment	Initiate payment with failed transaction	Invalid/failed payment	Donation should not be marked as successful	To be filled	Pass/Fail
TC019	Cancel payment	Start payment → Cancel transaction	Cancel payment	Transaction should be cancelled safely	To be filled	Pass/Fail
TC020	Generate donation receipt	Complete successful donation → View receipt	Successful transaction	Receipt should be generated with correct details	To be filled	Pass/Fail
TC021	Verify donation history	Login → Open Donation History	User with previous donations	Previous donation records should be displayed correctly	To be filled	Pass/Fail
TC022	Verify duplicate transaction handling	Submit the same payment request repeatedly	Same transaction	System should prevent duplicate donation records	To be filled	Pass/Fail

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9. Module 3 – Event Management

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC023	View available events	Open Events page	N/A	Available events should be displayed	To be filled	Pass/Fail
TC024	Create event as admin	Login as admin → Add Event → Enter details → Save	Temple Festival, 15-09-2026	Event should be created successfully	To be filled	Pass/Fail
TC025	Create event with missing fields	Leave required fields empty → Save	Blank event name/date	Validation messages should be displayed	To be filled	Pass/Fail
TC026	Enter invalid event date	Enter an invalid/past date	Past date	System should reject invalid event date where applicable	To be filled	Pass/Fail
TC027	Modify event	Admin → Select event → Edit → Save	Updated event details	Event information should be updated	To be filled	Pass/Fail
TC028	Delete event	Admin → Select event → Delete	Existing event	Event should be deleted after confirmation	To be filled	Pass/Fail
TC029	Register for event	User → Events → Select event → Register	Valid event	User should be registered successfully	To be filled	Pass/Fail
TC030	Duplicate event registration	Register for same event twice	Same user/event	System should prevent duplicate registration	To be filled	Pass/Fail
TC031	Event capacity limit	Register users until capacity is reached	Maximum capacity	System should stop additional registrations after capacity is reached	To be filled	Pass/Fail

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC032	Event registration confirmation	Complete registration	Valid registration	Confirmation should be displayed/sent	To be filled	Pass/Fail

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10. Module 4 – Admin Management

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC033	Admin login	Enter valid admin credentials	Admin credentials	Admin dashboard should be displayed	To be filled	Pass/Fail
TC034	Unauthorized admin access	Login as normal user → Try admin URL	Normal user account	Access should be denied	To be filled	Pass/Fail
TC035	View donation records	Admin → Donation Management	Existing donations	Donation records should be displayed	To be filled	Pass/Fail
TC036	Search donation record	Enter donor name/transaction ID	Hema / TXN001	Matching record should be displayed	To be filled	Pass/Fail
TC037	Filter donations	Select donation category/date	Annadanam	Relevant donation records should be displayed	To be filled	Pass/Fail
TC038	Manage users	Admin → User Management	Existing users	User information should be	To be filled	Pass/Fail

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
				displayed and manageable		
TC039	Generate donation report	Admin → Reports → Select date range	01-08-2026 to 25-08-2026	Correct report should be generated	To be filled	Pass/Fail
TC040	Generate event report	Admin → Reports → Events	Event records	Correct event report should be generated	To be filled	Pass/Fail

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11. Non-Functional Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Status
NFT001	Response time	Open major pages	Pages should load within acceptable time	To be filled	Pass/Fail
NFT002	Browser compatibility	Open system in Chrome, Edge and Firefox	System should function correctly	To be filled	Pass/Fail
NFT003	Usability	Navigate through registration, donation and events	Interface should be simple and understandable	To be filled	Pass/Fail
NFT004	Security	Attempt unauthorized admin access	Access should be denied	To be filled	Pass/Fail
NFT005	Session security	Logout and press browser back button	Protected pages should not be accessible without login	To be filled	Pass/Fail
NFT006	Data integrity	Complete a successful	Donation details should remain accurate	To be filled	Pass/Fail

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Status
		donation and check history			
NFT007	Error handling	Disconnect network during transaction	System should display appropriate error and avoid incorrect transaction status	To be filled	Pass/Fail
NFT008	Multiple users	Access system using multiple user accounts	System should handle users without data mixing	To be filled	Pass/Fail

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12. Boundary Value Analysis – Donation Amount

Assume the system accepts donations from **₹10 to ₹1,00,000**.

Test Data	Boundary Type	Expected Result
₹9	Below minimum	Reject
₹10	Minimum boundary	Accept
₹11	Just above minimum	Accept
₹99,999	Just below maximum	Accept
₹1,00,000	Maximum boundary	Accept
₹1,00,001	Above maximum	Reject
₹0	Invalid boundary	Reject
-₹10	Negative value	Reject

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13. Equivalence Partitioning

Donation Amount

Partition	Example	Expected Result
Valid amount	₹500	Accepted

Partition	Example	Expected Result
Minimum valid amount	₹10	Accepted
Below minimum	₹5	Rejected
Zero	₹0	Rejected
Negative	-₹100	Rejected
Above maximum	₹1,50,000	Rejected
Alphabetic	ABC	Rejected
Special characters	@#$	Rejected

Email Address

Partition	Example	Expected Result
Valid email	user@gmail.com	Accepted
Missing @	usergmail.com	Rejected
Missing domain	user@	Rejected
Empty email	Blank	Rejected
Invalid characters	user@@gmail.com	Rejected

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14. Decision Table – Donation Processing

Condition/Action	Case 1	Case 2	Case 3	Case 4
User logged in	Yes	Yes	Yes	No
Valid donation amount	Yes	Yes	No	Yes
Payment successful	Yes	No	-	-
Donation recorded	Yes	No	No	No
Receipt generated	Yes	No	No	No
Error message displayed	No	Yes	Yes	Yes

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15. State Transition Testing – Donation

The donation transaction should follow the following states:

Donation Initiated
↓
Amount and Category Entered
↓
Payment Processing
↓
Payment Successful
↓
Donation Recorded
↓
Receipt Generated

Alternative states:

Payment Processing → Payment Failed → Transaction Failed

Payment Processing → Payment Cancelled → Transaction Cancelled

Test cases should verify that the system does not incorrectly mark failed or cancelled payments as successful donations.

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16. Requirement Traceability Matrix

Requirement ID	**Covered Test Cases**
FR01 – Registration/Login	TC001–TC008
FR02 – Profile Management	TC001–TC004
FR04 – Donation Category	TC009–TC010
FR05 – Donation Amount	TC011–TC016
FR06 – Payment Processing	TC017–TC019
FR07 – Receipt Generation	TC020
FR08 – Donation History	TC021
FR09 – Event Management	TC023–TC028
FR10 – Event Registration	TC029–TC032

Requirement ID	Covered Test Cases
FR12 – Admin Dashboard	TC033–TC040
FR13 – User Management	TC038
FR14 – Donation Management	TC035–TC037
FR16 – Reports	TC039–TC040
FR18 – Logout	TC008

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17. Test Coverage

The test cases cover:

- **Functional testing –** Registration, login, donation, payment, receipt, events, administration and reports.
- **Boundary testing –** Minimum and maximum donation amounts.
- **Equivalence partitioning –** Valid and invalid input values.
- **Decision table testing –** Donation processing conditions.
- **State transition testing –** Payment and donation transaction states.
- **Negative testing –** Invalid login, invalid amounts, duplicate registration and failed payments.
- **Security testing –** Unauthorized admin access and session security.
- **Usability testing –** Navigation and ease of use.
- **Compatibility testing –** Multiple browsers.
- **Reliability testing –** Error handling and data consistency.

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18. Test Environment

Component	Description
Application Type	Web Application
Operating System	Windows/Linux
Browser	Google Chrome / Microsoft Edge / Mozilla Firefox

Component	Description
Database	MySQL / Firebase / MongoDB
Network	Internet/Wi-Fi
Testing Type	Manual Functional Testing
User Roles	Devotee/User and Administrator

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19. Entry Criteria

Testing can begin when:

- The application is available for testing.
- Required modules are developed.
- Test data is available.
- Test environment is configured.
- Login credentials are provided.
- Database and payment test environment are available.

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20. Exit Criteria

Testing can be completed when:

- All planned test cases have been executed.
- Critical defects are resolved.
- Major functional requirements are verified.
- Failed test cases are retested after fixes.
- Requirement coverage is achieved.
- No critical or high-severity defects remain open.

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21. Conclusion

The **Temple Donation and Event Management System** has been evaluated through comprehensive test scenarios covering functional and non-functional requirements.

The test cases include normal, boundary, invalid, exceptional, security, usability, and reliability conditions.

Testing techniques such as **Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing** are applied to improve test coverage and identify potential defects. Successful execution of these test cases will help ensure that the system provides accurate donation processing, reliable event management, secure user access, and a user-friendly experience.